

34.0.1 NIELIT Scientific Assistant June 2025 | Question: 30



30. Which chemical element is represented as "Hg" in the periodic table ?
(A) Silver (B) Tin (C) Copper (D) Mercury

30. Which chemical element is represented as " Hg " in the periodic table ?

- A. Silver B. Tin C. Copper D. Mercury

nielit-sta-2025

34.0.2 NIELIT Scientific Assistant June 2025 | Question: 29



29. If you are facing west and turn 90° clockwise, then 180° anti-clockwise, which direction will you be facing ?
(A) North (B) East (C) South (D) West

29. If you are facing west and turn 90° clockwise, then 180° anti-clockwise, which direction will you be facing ?

- (A) North
(B) East
(e) South
(D) West

nielit-sta-2025

34.0.3 NIELIT Scientific Assistant June 2025 | Question: 28



28. Maya is younger than Priya. Priya is elder than Rani but younger than Arjun. Arjun is elder than Komal. Who is the eldest ?
(A) Arjun (B) Maya (C) Priya (D) Rani

28. Maya is younger than Priya. Priya is elder than Rani but younger than Arjun. Arjun is elder than Komal. Who is the eldest ?

- A. Arjun B. Maya C. Priya D. Rani

nielit-sta-2025

34.0.4 NIELIT Scientific Assistant June 2025 | Question: 27



27. The simple interest on Rs. 1000 for 2 years at 5% per annum is :
(A) Rs. 50 (B) Rs. 100 (C) Rs. 75 (D) Rs. 60

27. The simple interest on Rs. 1000 for 2 years at 5% per annum is : 1

- A. Rs. 50 B. Rs. 100 C. Rs. 75 D. Rs. 60

nielit-sta-2025

34.0.5 NIELIT Scientific Assistant June 2025 | Question: 26



26. If a train travels 360 km at a speed of 90 km/h, how long does it take to complete the journey ?
(A) 5 hours (B) 4 hours (C) 3 hours (D) 6 hours

26. If a train travels 360 km at a speed of 90 km/h, how long does it take to complete

the journey ?

- A. 5 hours B. 4 hours C. 3 hours D. 6 hours

nielit-sta-2025

34.0.6 NIELIT Scientific Assistant June 2025 | Question: 25



25. The area of a triangle with base 10 cm and height 8 cm is :
(A) 30 sq cm (B) 40 sq cm (C) 60 sq cm (D) 80 sq cm

25. The area of a triangle with base 10 cm and height 8 cm is :

- A. 30 sq cm B. 40 sq cm C. 60 sq cm D. 80 sq cm

nielit-sta-2025

34.0.7 NIELIT Scientific Assistant June 2025 | Question: 24



24. The difference between compound interest and simple interest on Rs. 8000 at 5% per annum for 2 years is :
(A) Rs. 10 (B) Rs. 15 (C) Rs. 20 (D) Rs. 25

24. The difference between compound interest and simple interest on Rs. 8000 at 5% per annum for 2 years is:

- A. Rs. 10 B. Rs. 15 C. Rs. 20 D. Rs. 25

nielit-sta-2025

34.0.8 NIELIT Scientific Assistant June 2025 | Question: 23



23. If the perimeter of a rectangle is 40 cm and its length is 12 cm, find its breadth.
(A) 8 cm (B) 10 cm (C) 9 cm (D) 11 cm

23. If the perimeter of a rectangle is 40 cm and its length is 12 cm , find its breadth.

- A. 8 cm B. 10 cm C. 9 cm D. 11 cm

nielit-sta-2025

34.0.9 NIELIT Scientific Assistant June 2025 | Question: 22



22. Who won the 2024 Formula 1 Drivers' Championship ?
(A) George Russell (B) Max Verstappen
(C) Charles Leclerc (D) Sergio Pérez

22. Who won the 2024 Formula 1 Drivers' Championship ?

- A. George Russell B. Max Verstappen
C. Charles Leclerc D. Sergio Pérez

nielit-sta-2025

34.0.10 NIELIT Scientific Assistant June 2025 | Question: 21



21. Find the missing number: 12, 24, 48, 96, ?
(A) 190 (B) 192 (C) 194 (D) 196

21. Find the missing number: 12, 24, 48, 96 ?

- A. 190 B. 192 C. 194 D. 196

34.0.11 NIELIT Scientific Assistant June 2025 | Question: 20



20. If there are 8 teams in a tournament and each team plays with every other team once, how many matches are played ?
 (A) 36 (B) 30 (C) 32 (D) 28

20. If there are 8 teams in a tournament and each team plays with every other team once, how many matches are played ?

- A. 36 B. 30 C. 32 D. 28

34.0.12 NIELIT Scientific Assistant June 2025 | Question: 19



19. If the simple interest on a sum for 3 years at 10% per annum is Rs. 600, find the principal amount!
 (A) Rs. 3000 (B) Rs. 5000 (C) Rs. 2000 (D) Rs. 2500

19. If the simple interest on a sum for 3 years at 10% per annum is Rs. 600 , find the principal amount?

- A. Rs. 3000 B. Rs. 5000 C. Rs. 2000 D. Rs .2500

34.0.13 NIELIT Scientific Assistant June 2025 | Question: 18



18. Find the missing number :
 2, 6, 12, 20, ?
 (A) 28 (B) 30 (C) 32 (D) 40

18. Find the missing number: 2, 6, 12, 20, ?

- (A) 28
 (B) 30

- 32
 (D) 40

34.0.14 NIELIT Scientific Assistant June 2025 | Question: 17



17. Select the correct option to improve the sentence :
 He has been working here since four years.
 (A) working here for (B) working here from
 (C) working here while (D) No improvement

17. Select the correct option to improve the sentence:

He has been working here since four years.

- (A) working here for
 (B) working here from
 (C) working here while
 (B) No improvement

34.0.15 NIELIT Scientific Assistant June 2025 | Question: 16



16. What is the sum of the first 20 natural numbers ?
(A) 200 (B) 210 (C) 220 (D) 420

16. What is the sum of the first 20 natural numbers?

- A. 200 B. 210 C. 220 D. 420

nielit-sta-2025

34.0.16 NIELIT Scientific Assistant June 2025 | Question: 15



15. Nayab Saini was sworn in as the 11th Chief Minister of which State ?
(A) Punjab (B) Uttarakhand (C) Rajasthan (D) Haryana

15. Nayab Saini was sworn in as the 11th Chief Minister of which State?

- A. Punjab B. Uttarakhand C. Rajasthan D. Haryana

nielit-sta-2025

34.0.17 NIELIT Scientific Assistant June 2025 | Question: 14



14. Which pair is the odd one out ?
(A) Apple Fruit
(B) Dog Animal
(C) Rose Flower
(D) Car Road

14. Which pair is the odd one out?

- A. Apple Fruit B. Dog Animal C. Rose Flower D. Car Road

nielit-sta-2025

34.0.18 NIELIT Scientific Assistant June 2025 | Question: 13



13. A man walks 5 km north, then turns right and walks 3 km, then turns right again and walks 5 km. Which direction is he facing ?
(A) East (B) South (C) West (D) North

13. A man walks 5 km north, then turns right and walks 3 km , then turns right again and walks 5 km . Which direction is he facing ?

- (算) East
(B) South
(C) West
(D) North

nielit-sta-2025

34.0.19 NIELIT Scientific Assistant June 2025 | Question: 12



12. In a certain code language, LION is written as 50, then TIGER is written as _____
(A) 52 (B) 59 (C) 58 (D) 61

12. In a certain code language, LION is written as 50 , then TIGER is written as

- A. 52 B. 59 C. 58 D. 61

34.0.20 NIELIT Scientific Assistant June 2025 | Question: 11



11. In the given question, select the related word from the given alternatives.
Moon : Satellite :: Earth : ?
(A) Sun (B) Solar system (C) Planet (D) Round

11. In the given question, select the related word from the given alternatives.

Moon : Satellite :: Earth : ?

- A. Sun B. Solar system C. Planet D. Round

34.0.21 NIELIT Scientific Assistant June 2025 | Question: 10



10. Who wrote the novel "War and Peace" ?
(A) Anton Chekhov (B) Fyodor Dostoevsky
(C) Leo Tolstoy (D) Ivan Turgenev

10. Who wrote the novel "War and Peace" ?

- A. Anton Chekhov B. Fyodor Dostoevsky
C. Leo Tolstoy D. Ivan Turgenev

34.0.22 NIELIT Scientific Assistant June 2025 | Question: 9



9. What is the capital of Canada ?
(A) Toronto (B) Vancouver (C) Ottawa (D) Montreal

9. What is the capital of Canada ?

- A. Toronto B. Vancouver C. Ottawa D. Montreal

34.0.23 NIELIT Scientific Assistant June 2025 | Question: 8



8. Who was declared as the winner of the Pritzker Architecture Prize 2024 ?
(A) Riken Yamamoto (B) David Chipperfield
(C) B.V. Doshi (D) Yvonne Farrell

8. Who was declared as the winner of the Pritzker Architecture Prize 2024 ?

- A. Riken Yamamoto B. David Chipperfield
C. B.V. Doshi D. Yvonne Farrell

34.0.24 NIELIT Scientific Assistant June 2025 | Question: 7



7. Amit is elder than Rohit, but younger than Suman. Suman is younger than Neha. Rohit is elder than Kabir. Who is the youngest among them ?
(A) Rohit (B) Suman (C) Kabir (D) None of the options

7. Amit is elder than Rohit, but younger than Suman. Suman is younger than Neha. Rohit is elder than Kabir. Who is the youngest among them ?

A. Rohit

B. Suman

C. Kabir

D. None of the options

nielit-sta-2025

34.0.25 NIELIT Scientific Assistant June 2025 | Question: 6



6. Find the missing term :
BDF, CEG, DFH, ____, FIK
(A) EGI (B) EGH (C) EHI (D) DGI

6. Find the missing term :

BDF, CEG, DFH, FIK

A. EGI

B. EGH

C. EHI

D. DGI

nielit-sta-2025

34.0.26 NIELIT Scientific Assistant June 2025 | Question: 5



5. Find the next term in the series: 7, 14, 28, 56, ?
(A) 64 (B) 110 (C) 115 (D) 112

5. Find the next term in the series: 7, 14, 28, 56, 4

A. 64

B. 110

C. 115

D. 112

nielit-sta-2025

34.0.27 NIELIT Scientific Assistant June 2025 | Question: 4



4. Which is the largest lake in the World ?
(A) Caspian Sea (B) Baikal (C) Lake Superior (D) Ontario

4. Which is the largest lake in the World ?

A. Caspian Sea

B. Baikal

C. Lake Superior

D. Ontario

nielit-sta-2025

34.0.28 NIELIT Scientific Assistant June 2025 | Question: 3



3. What is the next term in the series: 3, 9, 27, 81, ?
(A) 162 (B) 243 (C) 99 (D) 150

3. What is the next term in the series: 3, 9, 27, 81, ?

A. 162

B. 243

C. 99

D. 150

nielit-sta-2025

34.0.29 NIELIT Scientific Assistant June 2025 | Question: 2



2. If $M * N + O$ means "M is the father of N and N is the mother of O," then how is M related to O ?
(A) Uncle (B) Father (C) Brother (D) Grandfather

2. If $M * N + O$ means " M is the father of N and N is the mother of O ," then how is M related to O ?

A. Uncle

B. Father

C. Brother

D. Grandfather

nielit-sta-2025

34.0.30 NIELIT Scientific Assistant June 2025 | Question: 1



1. A company sells a product of Rs. 1500 with a 20% discount and has no profit and no loss. If the company wants to make a 20% profit, at what price should the company sell that product ?
(A) Rs. 1400 (B) Rs. 1450 (C) Rs. 1440 (D) Rs. 1300

1. A company sells a product of Rs. 1500 with a 20% discount and has no profit and no loss. If the company wants to make a 20% profit, at what price should the company sell that product ?

- (A) Rs. 1400
- (B) Rs. 1450
- (G) Rs, 1440
- (D) Rs. 1300

nielit-sta-2025

34.0.31 NIELIT Scientific Assistant June 2025 | Question: 60



60. Which type of feasibility focuses on whether the hardware and software available can support the proposed system ?
(A) Operational Feasibility (B) Technical Feasibility
(C) Legal Feasibility (D) Economic Feasibility

60. Which type of feasibility focuses on whether the hardware and software available can support the proposed system ?

- A. Operational Feasibility
- B. Technical Feasibility
- C. Legal Feasibility
- D. Economic Feasibility

nielit-sta-2025

34.0.32 NIELIT Scientific Assistant June 2025 | Question: 59



59. Which of the following is the primary goal of information gathering in system development ?
(A) To design the system architecture
(B) To perform system testing
(C) To identify the available hardware and software
(D) To collect detailed user requirements and system needs

59. Which of the following is the primary goal of information gathering in system development ?

- (A) To design the system architecture
- B) To perform system testing
- (C) To identify the available hardware and software
- (D) To collect detailed user requirements and system needs

nielit-sta-2025

34.0.33 NIELIT Scientific Assistant June 2025 | Question: 58



58. Which of the following is the main purpose of the AM demodulator (envelope detector) ?
(A) To convert the modulated signal into the carrier frequency
(B) To filter out noise from the modulated signal
(C) To generate a new carrier signal
(D) To extract the message signal from the modulated signal

58. Which of the following is the main purpose of the AM demodulator (envelope detector) ?

- A. To convert the modulated signal into the carrier frequency
- B. To filter out noise from the modulated signal
- C. To generate a new carrier signal
- D. To extract the message signal from the modulated signal

nielit-sta-2025

34.0.34 NIELIT Scientific Assistant June 2025 | Question: 57



57. Consider a complete graph G with 4 vertices. The graph G has _____ spanning trees.
(A) 15 (B) 8 (C) 16 (D) 13

57. Consider a complete graph G with 4 vertices. The graph G has _____ spanning trees.

- A. 15
- B. 8
- C. 16
- D. 13

nielit-sta-2025

34.0.35 NIELIT Scientific Assistant June 2025 | Question: 56



56. The interval in which the function $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$ is strictly increasing or strictly decreasing is :
(A) Strictly increasing for $x \in \left[0, \frac{\pi}{4}\right] \cup \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
(B) Strictly increasing for $x \in (0, \pi)$ and strictly decreasing for $x \in (\pi, 2\pi)$
(C) Strictly increasing for $x \in \left[0, \frac{\pi}{4}\right] \cap \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
(D) Strictly decreasing for $x \in \left(0, \frac{\pi}{4}\right) \cup \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly increasing for $x \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

56. The interval in which the function $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$ is strictly increasing or strictly decreasing is :

- A. Strictly increasing for $x \in \left[0, \frac{\pi}{4}\right] \cup \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
- B. Strictly increasing for $x \in [0, \pi)$ and strictly decreasing for $x \in (\pi, 2\pi)$
- C. Strictly increasing for $x \in \left[0, \frac{\pi}{4}\right] \cap \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly decreasing for $x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
- D. Strictly decreasing for $x \in \left(0, \frac{\pi}{4}\right) \cup \left[\frac{5\pi}{4}, 2\pi\right]$ and strictly increasing for $x \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

nielit-sta-2025

34.0.36 NIELIT Scientific Assistant June 2025 | Question: 55



55. Which of the following is false in the case of a spanning tree of a graph G ?
(A) It is tree that spans G (B) It is a subgraph of the G
(C) It includes every vertex of the G (D) It can be either cyclic or acyclic

55. Which of the following is false in the case of a spanning tree of a graph G ?

- A. It is tree that spans G
- B. It is a subgraph of the G
- C. It includes every vertex of the G
- D. It can be either cyclic or acyclic

nielit-sta-2025

34.0.37 NIELIT Scientific Assistant June 2025 | Question: 54



54. A bag contains ' x ' red, ' $x+5$ ' blue and ' $x+7$ ' grey balls. If two balls are randomly drawn from the bag and the probability that both the balls are of same colour is $\frac{148}{435}$, then the total number of balls in the bag is :
(A) 36 (B) 64 (C) 58 (D) 30

54. A bag contains ' x ' red, ' $x + 5$ ' blue and ' $x + 7$ ' grey balls. If two balls are randomly drawn from the bag and the probability that both the balls are of same colour is $\frac{148}{435}$, then the total number of balls in the bag is :

- A. 36 B. 64 C. 58 D. 30

nielit-sta-2025

34.0.38 NIELIT Scientific Assistant June 2025 | Question: 53



53. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\tan 2x}{x - \frac{\pi}{2}} \right) =$ _____
 (A) 0 (B) 1 (C) 2 (D) Undefined

53. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\tan 2x}{x - \frac{\pi}{2}} \right) =$

- A. 0 B. 1 C. 2 D. Undefined

nielit-sta-2025

34.0.39 NIELIT Scientific Assistant June 2025 | Question: 52



52. If $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$ is an eigen vector of $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, the eigen value corresponding to the given eigen vector is :
 (A) 1 (B) 4 (C) 6 (D) -4.5

52. If $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$ is an eigen vector of $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, the eigen value corresponding to the given eigen vector is:

- A. 1 B. 4 C. 6 D. -4.5

nielit-sta-2025

34.0.40 NIELIT Scientific Assistant June 2025 | Question: 51



51. The approximate value of integral $\int_0^1 (3x+7)dx$ by trapezoidal rule of the step size 0.0001 is
 (A) $\frac{3}{2}$ (B) 7 (C) $\frac{17}{2}$ (D) None of the options

51. The approximate value of integral $\int_0^1 (3x + 7)dx$ by trapezoidal rule of the step size 0.0001 is

- A. $\frac{3}{2}$ B. 7 C. $\frac{17}{2}$ D. None of the options

nielit-sta-2025

34.0.41 NIELIT Scientific Assistant June 2025 | Question: 50



50. Which of the following is simplified sum of product form for the Boolean expression $(P + Q' + R')(P + Q' + R)(P + Q + R)$?
 (A) $P + QR'$ (B) $P + Q'R$ (C) $P + Q'R'$ (D) $P' + Q'R'$

50. Which of the following is simplified sum of product form for the Boolean expression ?
 $(P + Q' + R') (P + Q' + R) (P + Q + R')$

- A. $P + QR'$
 C. $P + Q'R'$

- B. $P + Q'R$
 D. $P' + Q'R'$

nielit-sta-2025

34.0.42 NIELIT Scientific Assistant June 2025 | Question: 49



49. If $A = \text{Baby is eating}$, $B = \text{Baby is hungry}$, then the correct propositional logic to the English sentence "Baby is eating and not hungry" is :
 (A) $A \Rightarrow \neg B$ (B) $A \wedge \neg B$ (C) $\neg(A \Rightarrow \neg B)$ (D) $A \vee \neg B$

49. If $A = \text{Baby is eating}$, $B = \text{Baby is hungry}$, then the correct propositional logic to the English sentence

- A. $A \Rightarrow \neg B$ and not hungry * is :
 C. $\neg(A \Rightarrow \neg B)$

- B. $A \wedge \neg B$
 D. $A \vee \neg B$

nielit-sta-2025

34.0.43 NIELIT Scientific Assistant June 2025 | Question: 48



48. For what values of x :

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$$

 (A) 1 (B) 2 (C) 3 (D) -1

48. For what values of x :

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$$

- A. 1 B. 2 C. 3 D. -1

nielit-sta-2025

34.0.44 NIELIT Scientific Assistant June 2025 | Question: 47



47. For two events A and B , we have the following probabilities.
 $P(A) = P\left(\frac{A}{B}\right) = \frac{1}{4}$, $P\left(\frac{B}{A}\right) = \frac{1}{2}$
 Consider the statements :
 (1) A and B are independent. (2) $P(B) = \frac{1}{2}$
 Which of the following is correct ?
 (A) (1) is True; (2) is False (B) (1) is True; (2) is True
 (C) (1) is False; (2) is False (D) (1) is False; (2) is True

47. For two events A and B , we have the following probabilities.

$$P(A) = P\left(\frac{A}{B}\right) = \frac{1}{4}, P\left(\frac{B}{A}\right) = \frac{1}{2}$$

Consider the statements :

(1) A and B are independent.

(2) $P(B) = \frac{1}{2}$

Which of the following is correct ?

A. (1) is True; (2) is False

B. (1) is True; (2) is True

C. (1) is False; (2) is False

D. (1) is False; (2) is True

nielit-sta-2025

34.0.45 NIELIT Scientific Assistant June 2025 | Question: 46



46. The sum of all the 4 digit numbers that can be formed with the digits 3, 4, 5 and 6 is :
(A) 119988 (B) 11988
(C) 191988 (D) None of the options

46. The sum of all the 4 digit numbers that can be formed with the digits 3, 4, 5 and 6 is :

A. 119988

B. 11988

C. 191988

D. None of the options

nielit-sta-2025

34.0.46 NIELIT Scientific Assistant June 2025 | Question: 45



45. Let $X = \{1, 2, 3, 4\}$. If
 $A = \{\langle x, y \rangle \mid x \in X \wedge y \in X \wedge (x - y) \text{ is a nonzero multiple of } 2\}$
 $B = \{\langle x, y \rangle \mid x \in X \wedge y \in X \wedge (x - y) \text{ is a nonzero multiple of } 3\}$
Then $A \cap B$ is :
(A) $\{(1, 4), (4, 1)\}$
(B) $\{(1, 3), (3, 1), (2, 4), (4, 2)\}$
(C) $\{(1, 3), (3, 1), (2, 4), (4, 2), (1, 4), (4, 1)\}$
(D) ϕ

45. Let $X = \{1, 2, 3, 4\}$. If

$A = \{\langle x, y \rangle \mid x \in X \wedge y \in X \wedge (x - y) \text{ is a nonzero multiple of } 2\}$

$B = \{\langle x, y \rangle \mid x \in X \wedge y \in X \wedge (x - y) \text{ is a nonzero multiple of } 3\}$ Then $A \cap B$ is :

A. $\{(1, 4), (4, 1)\}$

B. $\{(1, 3), (3, 1), (2, 4), (4, 2)\}$

C. $\{(1, 3), (3, 1), (2, 4), (4, 2), (1, 4), (4, 1)\}$

D. ϕ

nielit-sta-2025

34.0.47 NIELIT Scientific Assistant June 2025 | Question: 44



44. The LU factorization requires :
(A) $\frac{n^3}{6} - \frac{n}{5}$ multiplication/division
(B) $\frac{n^3}{3} - \frac{n^2}{2} + \frac{n}{6}$ addition/subtraction
(C) $\frac{n^3}{5} - \frac{n^2}{3} + \frac{n}{6}$ addition/subtraction
(D) None of the above

44. The LU factorization requires:

A. $\frac{n^3}{6} - \frac{n}{5}$ multiplication/division

B. $\frac{n^3}{3} - \frac{n^2}{2} + \frac{n}{6}$ addition/subtraction

C. $\frac{n^3}{5} - \frac{n^2}{3} + \frac{n}{6}$ addition/subtraction

D. None of the above

nielit-sta-2025

34.0.48 NIELIT Scientific Assistant June 2025 | Question: 43

43. The number of straight lines that can be drawn out of 12 points of which 8 are collinear is.
(A) 39 (B) 29 (C) 49 (D) 59

43. The number of straight lines that can be drawn out of 12 points of which 8 are collinear is:

- A. 39 B. 29 C. 49 D. 59

nielit-sta-2025

34.0.49 NIELIT Scientific Assistant June 2025 | Question: 42

42. The product of two numbers is 120, and their sum is 22. What are the numbers ?
(A) 12, 11 (B) 6, 20 (C) 8, 15 (D) 10, 12

42. The product of two numbers is 120 , and their sum is 22 . What are the numbers ?

- A. 12,11 B. 6,20 C. 8,15 D. 10,12

nielit-sta-2025

34.0.50 NIELIT Scientific Assistant June 2025 | Question: 41

41. First Bank in India to introduce talking ATMs for disabled persons :
(A) SBI (B) Punjab National Bank
(C) ICICI Bank (D) Union Bank of India

41. First Bank in India to introduce talking ATMs for disabled persons :

- A. SBI B. Punjab National Bank
C. ICICI Bank D. Union Bank of India

nielit-sta-2025

34.0.51 NIELIT Scientific Assistant June 2025 | Question: 40

40. If the diameter of a circle is reduced by 50%, what happens to its area ?
(A) Reduced by 25% (B) Reduced by 50%
(C) Reduced by 75% (D) Reduced by 100%

40. If the diameter of a circle is reduced by 50%, what happens to its area ?

- A. Reduced by 25% B. Reduced by 50%
C. Reduced by 75% D. Reduced by 100%

nielit-sta-2025

34.0.52 NIELIT Scientific Assistant June 2025 | Question: 39

39. What is the probability of rolling a prime number on a fair six-sided die ?
(A) 1/3 (B) 1/5 (C) 1/2 (D) 1/6

39. What is the probability of rolling a prime number on a fair six-sided die ?

- A. 1/3 B. 1/5 C. 1/2 D. 1/6

nielit-sta-2025

Answer key

34.0.53 NIELIT Scientific Assistant June 2025 | Question: 38



38. A person walks 10 km north, then turns right and walks 2 km, then turns right again and walks 10 km. Finally, he turns left and walks 4 km. How far is he from his starting point ?

- (A) 6 km (B) 4 km (C) 10 km (D) 8 km

38. A person walks 10 km north, then turns right and walks 2 km, then turns right again and walks 10 km. Finally, he turns left and walks 4 km. How far is he from his starting point ?

- A. 6 km B. 4 km C. 10 km D. 8 km

nielit-sta-2025

34.0.54 NIELIT Scientific Assistant June 2025 | Question: 37



37. Who was the Governor of Reserve Bank of India as on 5th March 2025 ?

- (A) Sanjay Malhotra (B) Shaktikant Das
(C) S. Jaishankar (D) R.N. Malhotra

37. Who was the Governor of Reserve Bank of India as on 5th March 2025 ?

- A. Sanjay Malhotra B. Shaktikant Das
C. S. Jaishankar D. R.N. Malhotra

nielit-sta-2025

34.0.55 NIELIT Scientific Assistant June 2025 | Question: 36



36. If a and b are positive integers, such that $a^*a = 4$ $b^*b = 36$, then what is a^*b ?

- (A) 19 (B) 12 (C) 10 (D) 20

36. If a and b are positive integers, such that $a^*a = 4$ $b^*b = 36$, then what is a^*b ?

- A. 19 B. 12 C. 10 D. 20

nielit-sta-2025

Answer key

34.0.56 NIELIT Scientific Assistant June 2025 | Question: 35



35. The new manager is known for his _____ nature.

- (A) Ambiguous (B) Animosity (C) Hostile (D) Amiable

35. The new manager is known for his _____ nature.

- A. Ambiguous B. Animosity C. Hostile D. Amiable

nielit-sta-2025

34.0.57 NIELIT Scientific Assistant June 2025 | Question: 34



34. Study of insects is known as _____.

- (A) Astrology (B) Emetology (C) Entomology (D) Geology

34. Study of insects is known as _____.

- A. Astrology B. Emetology C. Entomology D. Geology

Answer key

34.0.58 NIELIT Scientific Assistant June 2025 | Question: 33



33. If STAR is coded as VWDU, how is MOON coded ?
 (A) PRRQ (B) QPQP (C) ORRP (D) PPRQ

33. If STAR is coded as VWDU, how is MOON coded ?

- A. PRRQ B. QPQP C. ORRP D. PPRQ

Answer key

34.0.59 NIELIT Scientific Assistant June 2025 | Question: 32



32. Find the next term :
 AZ, BY, CX, ?
 (A) DW (B) EV (C) DU (D) FW

32. Find the next term :

AZ, BY, CX, ?

- A. DW B. EV C. DU D. FW

Answer key

34.0.60 NIELIT Scientific Assistant June 2025 | Question: 31



31. Statements :
 All cats are dogs.
 Some dogs are lions.
 Conclusions :
 I. Some lions are cats.
 II. All lions are dogs.
 (A) Only Conclusion I follows (B) Only Conclusion II follows
 (C) Both follow (D) Neither follows

31. Statements :

All cats are dogs.

Some dogs are lions.

Conclusions:

I. Some lions are cats.

II. All lions are dogs.

(A) Only Conclusion I follows

(C) Both follow

(B) Only Conclusion II follows

(D). Neither follows

34.0.61 NIELIT Scientific Assistant June 2025 | Question: 90



90. Which OSI layer is responsible for data compression and encryption ?
(A) Transport (B) Application (C) Presentation (D) Session

90. Which OSI layer is responsible for data compression and encryption ?

- A. Transport B. Application C. Presentation D. Session

nielit-sta-2025

34.0.62 NIELIT Scientific Assistant June 2025 | Question: 89



89. Which normal form deals with removing partial dependencies ?
(A) 1st Normal Form (1NF)
(B) 2nd Normal Form (2NF)
(C) 3rd Normal Form (3NF)
(D) Boyce-Codd Normal Form (BCNF)

89. Which normal form deals with removing partial dependencies ?

- A. 1st Normal Form (1NF) B. 2nd Normal Form (2NF)
C. 3rd Normal Form (3NF) D. Boyce-Codd Normal Form (BCNF)

nielit-sta-2025

34.0.63 NIELIT Scientific Assistant June 2025 | Question: 88



88. Which of the following is true for the LL(1) parsing method ?
(A) It uses left recursion in its grammar
(B) It can be used for all context-free languages
(C) It requires a look-ahead of 1 symbol
(D) It is a type of shift-reduce parsing

88. Which of the following is true for the LL(1) parsing method?

- (A) It uses left recursion in its grammar
(B) It can be used for all context-free languages
(C) It requires a look-ahead of 1 symbol
(D) It is a type of shift-reduce parsing

nielit-sta-2025

34.0.64 NIELIT Scientific Assistant June 2025 | Question: 87



87. What is the main purpose of a device driver ?
(A) To control the physical hardware of the device
(B) To manage network communication
(C) To control memory allocation
(D) To execute system calls

87. What is the main purpose of a device driver ?

- A. To control the physical hardware of the device B. To manage network communication
C. To control memory allocation D. To execute system calls

nielit-sta-2025

34.0.65 NIELIT Scientific Assistant June 2025 | Question: 86



86. What does a Push-Down Automaton (PDA) use to make decisions ?
(A) Stack (B) Queue (C) Tape (D) Register

86. What does a Push-Down Automaton (PDA) use to make decisions ?

A. Stack

B. Queue

C. Tape

D. Register

nielit-sta-2025

34.0.66 NIELIT Scientific Assistant June 2025 | Question: 85



85. Which of the following is a feature of a Real-Time Operating System (RTOS) ?
- (A) Processes are executed in a sequential manner.
 - (B) Tasks are executed without strict timing constraints.
 - (C) It guarantees that tasks will be executed within a predefined time limit.
 - (D) It does not support multitasking.

85. Which of the following is a feature of a Real-Time Operating System (RTOS) ?

- A. Processes are executed in a sequential manner.
- B. Tasks are executed without strict timing constraints.
- C. It guarantees that tasks will be executed within a predefined time limit.
- D. It does not support multitasking.

nielit-sta-2025

34.0.67 NIELIT Scientific Assistant June 2025 | Question: 84



84. What does the size of a Page Table Entry (PTE) depend on in a paging system ?
- (A) The number of bits required to represent the frame number
 - (B) The size of the virtual memory address
 - (C) The size of the physical memory address
 - (D) The size of the page

84. What does the size of a Page Table Entry (PTE) depend on in a paging system ?

- A. The number of bits required to represent the frame number
- B. The size of the virtual memory address
- C. The size of the physical memory address
- D. The size of the page

nielit-sta-2025

34.0.68 NIELIT Scientific Assistant June 2025 | Question: 83



83. Which of the following is an example of passing a parameter by reference in C ?
- (A) void func(int x)
 - (B) void func(int &x)
 - (C) void func(int x, int y)
 - (D) void func(int *x)

83. Which of the following is an example of passing a parameter by reference in C ?

- (A) void func(int x)
- (C) void func(int x , int y)
- (B) void func(int $\&x$)
- (D) void func(int $*x$)

nielit-sta-2025

34.0.69 NIELIT Scientific Assistant June 2025 | Question: 82



82. The modulation index of AM is defined as :
- (A) The ratio of carrier frequency to message signal frequency
 - (B) The ratio of the peak amplitude of the modulated signal to the peak amplitude of the carrier
 - (C) The ratio of the carrier power to the total power in the AM signal
 - (D) The ratio of the amplitude of the modulating signal to the amplitude of the carrier signal

82. The modulation index of AM is defined as :

- (A) The ratio of carrier frequency to message signal frequency
- (B) The ratio of the peak amplitude of the modulated signal to the peak amplitude of the carrier
- (D) The ratio of the carrier power to the total power in the AM signal

nielit-sta-2025

34.0.70 NIELIT Scientific Assistant June 2025 | Question: 81



81. If the input to an LTI system is a random signal with a known mean and variance, the output will have :
- (A) The same mean and variance as the input signal
 - (B) The same mean, but a different variance
 - (C) The same variance, but a different mean
 - (D) A completely random mean and variance

81. If the input to an LTI system is a random signal with a known mean and variance, the output will have:

- A. The same mean and variance as the input signal
- B. The same mean, but a different variance
- C. The same variance, but a different mean
- D. A completely random mean and variance

nielit-sta-2025

34.0.71 NIELIT Scientific Assistant June 2025 | Question: 80



80. What is the purpose of public key cryptography ?
- (A) To encrypt data using a single shared key
 - (B) To replace the need for encryption in secure communication
 - (C) To ensure confidentiality by using two keys: a public key and a private key
 - (D) To decrypt digital signatures

80. What is the purpose of public key cryptography ?

- A. To encrypt data using a single shared key
- B. To replace the need for encryption in secure communication
- C. To ensure confidentiality by using two keys: a public key and a private key
- D. To decrypt digital signatures

nielit-sta-2025

34.0.72 NIELIT Scientific Assistant June 2025 | Question: 79



79. Which isolation level in database systems allows transactions to be executed concurrently but without allowing dirty reads ?
- (A) Read Uncommitted
 - (B) Read Committed
 - (C) Repeatable Read
 - (D) Serializable

79. Which isolation level in database systems allows transactions to be executed concurrently but without allowing dirty reads ?

- (A) Read Uncommitted
- (C) Repeatable Read
- (B) Read Committed
- (D) Serializable

nielit-sta-2025

34.0.73 NIELIT Scientific Assistant June 2025 | Question: 78



78. What does the term "shift-reduce conflict" refer to in a parser ?
 (A) When a loop is detected during parsing
 (B) When the grammar has ambiguous rules
 (C) When there is an error in semantic analysis
 (D) When a parser can't decide whether to shift or reduce based on the current state

78. What does the term "shift-reduce conflict" refer to in a parser ?
- (A) When a loop is detected during parsing
 - (B) When the grammar has ambiguous rules
 - (C) When there is an error in semantic analysis
 - (B) When a parser can't decide whether to shift or reduce based on the current state

nielit-sta-2025

34.0.74 NIELIT Scientific Assistant June 2025 | Question: 77



77. Which method is commonly used for converting a decimal number into its binary equivalent in a floating-point representation ?
 (A) Successive division by 2
 (B) Successive subtraction of 1
 (C) Direct conversion from decimal to binary
 (D) Multiplying by 10

77. Which method is commonly used for converting a decimal number into its binary equivalent in a floating-point representation ?

- | | |
|---|--------------------------------|
| A. Successive division by 2 | B. Successive subtraction of 1 |
| C. Direct conversion from decimal to binary | D. Multiplying by 10 |

nielit-sta-2025

34.0.75 NIELIT Scientific Assistant June 2025 | Question: 76



76. Which of the following is the correct asymptotic notation for an algorithm that grows slower than $O(n^2)$ but faster than $O(n)$?
 (A) $O(n \log n)$ (B) $O(n^2)$ (C) $O(n^3)$ (D) $O(\log n)$

76. Which of the following is the correct asymptotic notation for an algorithm that grows slower than $O(n^2)$ but faster than $O(n)$?

- | | |
|------------------|----------------|
| A. $O(n \log n)$ | B. $O(n^2)$ |
| C. $O(n^3)$ | D. $O(\log n)$ |

nielit-sta-2025

34.0.76 NIELIT Scientific Assistant June 2025 | Question: 75



75. Which of the following addressing modes is used when the operand is directly specified in the instruction ?
 (A) Immediate Addressing (B) Register Addressing
 (C) Direct Addressing (D) Indirect Addressing

75. Which of the following addressing modes is used when the operand is directly

specified in the instruction?

- (A) Immediate Addressing
- (C) Direct Addressing
- (B) Register Addressing
- (D) Indirect Addressing

nielit-sta-2025

34.0.77 NIELIT Scientific Assistant June 2025 | Question: 74



74. What is the advantage of using a full adder over a half adder in binary addition ?
(A) A full adder does not require a carry-in input.
(B) A full adder can add two 4-bit numbers.
(C) A full adder handles carry inputs as well as carry outputs.
(D) A full adder is used in sequential circuits only.

74. What is the advantage of using a full adder over a half adder in binary addition ?

- A. A full adder does not require a carry-in input.
- B. A full adder can add two 4 -bit numbers.
- C. A full adder handles carry inputs as well as carry outputs.
- D. A full adder is used in sequential circuits only.

nielit-sta-2025

34.0.78 NIELIT Scientific Assistant June 2025 | Question: 73



73. Which of the following phases in the Software Development Life Cycle (SDLC) involves gathering and analyzing user requirements ?
(A) Design (B) Implementation
(C) Requirement Analysis (D) Maintenance

73. Which of the following phases in the Software Development Life Cycle (SDLC) involves gathering and analyzing user requirements ?

- A. Design
- B. Implementation
- C. Requirement Analysis
- D. Maintenance

nielit-sta-2025

34.0.79 NIELIT Scientific Assistant June 2025 | Question: 72



72. Which of the following problems can be solved using a greedy approach ?
(A) Knapsack problem (0/1 version)
(B) Job Scheduling with deadlines and profits
(C) Fibonacci sequence calculation
(D) Finding the shortest path in a weighted graph

72. Which of the following problems can be solved using a greedy approach ?

- A. Knapsack problem (0/1 version)
- B. Job Scheduling with deadlines and profits
- C. Fibonacci sequence calculation
- D. Finding the shortest path in a weighted graph

nielit-sta-2025

34.0.80 NIELIT Scientific Assistant June 2025 | Question: 71



71. What is the purpose of the sliding window protocol in data communication ?
- (A) To manage the amount of data that can be sent before receiving an acknowledgment
 - (B) To check for errors in received data
 - (C) To determine the routing path in a network
 - (D) To initiate a connection between devices

71. What is the purpose of the sliding window protocol in data communication ?

- A. To manage the amount of data that can be sent before receiving an acknowledgment
- B. To check for errors in received data
- C. To determine the routing path in a network
- D. To initiate a connection between devices

nielit-sta-2025

34.0.81 NIELIT Scientific Assistant June 2025 | Question: 70



70. What is a key advantage of B+ trees over B- trees ?
- (A) B+ trees allow multiple key values per node
 - (B) B+ trees use more disk space
 - (C) B+ trees store data at both internal and leaf nodes
 - (D) B+ trees have better performance for range queries

70. What is a key advantage of B + trees over B - trees ?

- A. B+ trees allow multiple key values per node
- B. B+ trees use more disk space
- C. B+ trees store data at both internal and leaf nodes
- D. B+ trees have better performance for range queries

nielit-sta-2025

34.0.82 NIELIT Scientific Assistant June 2025 | Question: 69



69. Which traversal method can be used to print the nodes of a binary tree in ascending order ?
- (A) Pre-order traversal
 - (B) Post-order traversal
 - (C) In-order traversal
 - (D) Level-order traversal

69. Which traversal method can be used to print the nodes of a binary tree in ascending order ?

- (A) Pre-order traversal
- (B) Post-order traversal
- (c) In-order traversal
- (D) Level-order traversal

nielit-sta-2025

34.0.83 NIELIT Scientific Assistant June 2025 | Question: 68



68. Which of the following describes binding in C ?
- (A) The process of assigning a value to a variable
 - (B) The process of declaring the type of a variable
 - (C) The process of associating a function with a return type
 - (D) The process of associating a function name with a memory address

68. Which of the following describes binding in C ?

- A. The process of assigning a value to a variable

- B. The process of declaring the type of a variable
- C. The process of associating a function with a return type
- D. The process of associating a function name with a memory address

nielit-sta-2025

Answer key 

34.0.84 NIELIT Scientific Assistant June 2025 | Question: 67



67. Which of the following algorithms is used to find the minimum spanning tree in a graph ?
(A) Dijkstra's Algorithm (B) Kruskal's Algorithm
(C) Bellman-Ford Algorithm (D) Floyd-Warshall Algorithm

67. Which of the following algorithms is used to find the minimum spanning tree in a graph ?

- (A) Dijkstra's Algorithm
- (C) Bellman-Ford Algorithm
- (B) Kruskal's Algorithm
- (D) Floyd-Warshall Algorithm

nielit-sta-2025

34.0.85 NIELIT Scientific Assistant June 2025 | Question: 66



66. In which addressing mode, the address of the operand is given by the sum of a constant value and the content of a register ?
(A) Indexed Addressing (B) Base Register Addressing
(C) Direct Addressing (D) Register Indirect Addressing

66. In which addressing mode, the address of the operand is given by the sum of a constant value and the content of a register ?

- (A) Indexed Addressing
- (C) Direct Addressing
- (B) Base Register Addressing
- (D) Register Indirect Addressing

nielit-sta-2025

Answer key 

34.0.86 NIELIT Scientific Assistant June 2025 | Question: 65



65. In PCM, the quantization error is caused by :
(A) Sampling the signal
(B) The difference between the input signal and the quantized value
(C) The modulation of the signal
(D) The carrier frequency variation

65. In PCM, the quantization error is caused by :

- A. Sampling the signal
- B. The difference between the input signal and the quantized value
- C. The modulation of the signal
- D. The carrier frequency variation

nielit-sta-2025

34.0.87 NIELIT Scientific Assistant June 2025 | Question: 64

64. What is the key difference between a DFA (Deterministic Finite Automaton) and an NFA (Nondeterministic Finite Automaton) ?
- (A) A DFA has a single state transition for each symbol, while an NFA can have multiple transitions.
 - (B) An NFA has a single state transition for each symbol, while a DFA can have multiple transitions.
 - (C) DFAs are less powerful than NFAs.
 - (D) NFAs can recognize more languages than DFAs.

64. What is the key difference between a DFA (Deterministic Finite Automaton) and an NFA (Nondeterministic Finite Automaton) ?

- A. A DFA has a single state transition for each symbol, while an NFA can have multiple transitions.
- B. An NFA has a single state transition for each symbol, while a DFA can have multiple transitions.
- C. DFAs are less powerful than NFAs.
- D. NFAs can recognize more languages than DFAs.

nielit-sta-2025

34.0.88 NIELIT Scientific Assistant June 2025 | Question: 63

63. What is the time complexity for searching an element in a balanced Binary Search Tree ?
- (A) $O(1)$
 - (B) $O(\log n)$
 - (C) $O(n)$
 - (D) $O(n \log n)$

63. What is the time complexity for searching an element in a balanced Binary Search Tree ?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$

nielit-sta-2025

34.0.89 NIELIT Scientific Assistant June 2025 | Question: 62

62. Which of the following is the primary operation for adding an element to a queue in C ?
- (A) Enqueue
 - (B) Dequeue
 - (C) Push
 - (D) Pop

62. Which of the following is the primary operation for adding an element to a queue in C ?

- A. Enqueue
- B. Dequeue
- C. Push
- D. Pop

nielit-sta-2025

34.0.90 NIELIT Scientific Assistant June 2025 | Question: 61

61. What is the main purpose of the synthesis of combinational circuits ?
- (A) To optimize the circuit speed
 - (B) To find the minimal sum of products
 - (C) To design the flip-flops for sequential circuits
 - (D) To convert the truth table into a Boolean expression

61. What is the main purpose of the synthesis of combinational circuits ?

- (A) To optimize the circuit speed
- B) To find the minimal sum of products
- (C) To design the flip-flops for sequential circuits
- (D) To convert the truth table into a Boolean expression

34.0.91 NIELIT Scientific Assistant June 2025 | Question: 120

20. Which of the following is the main characteristic of sequential files ?

- (A) Data is stored in an unordered fashion
- (B) Data is stored in sequential order
- (C) Data is indexed for fast access
- (D) Data can be accessed randomly

20. Which of the following is the main characteristic of sequential files?

- A. Data is stored in an unordered fashion
- B. Data is stored in sequential order
- C. Data is indexed for fast access
- D. Data can be accessed randomly

nielit-sta-2025

34.0.92 NIELIT Scientific Assistant June 2025 | Question: 119

19. Which of the following problems is solved using dynamic programming ?

- (A) Merge sort
- (B) Depth-first search
- (C) Breadth-first search
- (D) Fibonacci sequence calculation

19. Which of the following problems is solved using dynamic programming ?

- A. Merge sort
- B. Depth-first search
- C. Breadth-first search
- D. Fibonacci sequence calculation

nielit-sta-2025

34.0.93 NIELIT Scientific Assistant June 2025 | Question: 118

8. Which of the following is true about 'Level 0' DFD (Context Diagram) ?

- (A) It shows the complete system with all its processes and data flows
- (B) It shows a high-level view of the system with one process and its interactions with external entities
- (C) It is used to detail the internal components of the system
- (D) It shows only data stores and external entities

18. Which of the following is true about 'Level 0' DFD (Context Diagram) ?

- A. It shows the complete system with all its processes and data flows
- B. It shows a high-level view of the system with one process and its interactions with external entities
- C. It is used to detail the internal components of the system
- D. It shows only data stores and external entities

nielit-sta-2025

34.0.94 NIELIT Scientific Assistant June 2025 | Question: 117

17. Which data structure is used by the lexical analyzer in a compiler ?

- (A) Stack
- (B) Queue
- (C) Symbol table
- (D) Tree

17. Which data structure is used by the lexical analyzer in a compiler ?

- A. Stack B. Queue C. Symbol table D. Tree

nielit-sta-2025

34.0.95 NIELIT Scientific Assistant June 2025 | Question: 116



6. Which of the following is the most powerful computational model ?
(A) Finite Automaton (B) Push-Down Automaton
(C) Turing Machine (D) Linear Bounded Automaton

6. Which of the following is the most powerful computational model ?
(A) Finite Automaton
(C) Turing Machine
(B) Push-Down Automaton
(D) Linear Bounded Automaton

nielit-sta-2025

34.0.96 NIELIT Scientific Assistant June 2025 | Question: 115



115. In CPU control design, which of the following is responsible for generating control signals ?
(A) ALU (B) Control Unit (C) Registers (D) Data Bus

115. In CPU control design, which of the following is responsible for generating control signals ?

- A. ALU B. Control Unit C. Registers D. Data Bus

nielit-sta-2025

34.0.97 NIELIT Scientific Assistant June 2025 | Question: 114



114. What is the primary disadvantage of using indirect addressing mode ?
(A) It eliminates the need for instructions
(B) It is faster than direct addressing
(C) It reduces the need for registers
(D) It requires extra memory access

114. What is the primary disadvantage of using indirect addressing mode ?

- A. It eliminates the need for instructions B. It is faster than direct addressing
C. It reduces the need for registers D. It requires extra memory access

nielit-sta-2025

34.0.98 NIELIT Scientific Assistant June 2025 | Question: 113



113. What is the main advantage of using a DFA over an NFA ?
(A) DFAs are easier to construct.
(B) DFAs always have fewer states than NFAs.
(C) DFAs have a deterministic behavior, making them more predictable.
(D) DFAs can recognize more complex languages than NFAs.

113. What is the main advantage of using a DFA over an NFA ?

- A. DFAs are easier to construct.
B. DFAs always have fewer states than NFAs.
C. DFAs have a deterministic behavior, making them more predictable.
D. DFAs can recognize more complex languages than NFAs.

34.0.99 NIELIT Scientific Assistant June 2025 | Question: 112



112. Which of the following best describes the purpose of the "backpatching" technique in code generation ?

(A) It is used to resolve forward jumps by filling in missing addresses.

(B) It is used to optimize code by evaluating constant expressions at compile-time.

(C) It is used to detect and correct syntax errors in the source code.

(D) It is used for constant folding.

112. Which of the following best describes the purpose of the "backpatching" technique in code generation?

- A. It is used to resolve forward jumps by filling in missing addresses.
- B. It is used to optimize code by evaluating constant expressions at compile-time.
- C. It is used to detect and correct syntax errors in the source code.
- D. It is used for constant folding.

34.0.100 NIELIT Scientific Assistant June 2025 | Question: 111



111. Which of the following is the primary objective of the "round-robin" CPU scheduling algorithm ?

(A) Minimize the average waiting time

(B) Avoid starvation

(C) Minimize the turnaround time

(D) Provide each process with a fair share of CPU time

111. Which of the following is the primary objective of the "round-robin" CPU scheduling algorithm ?

- A. Minimize the average waiting time
- B. Avoid starvation
- C. Minimize the turnaround time
- D. Provide each process with a fair share of CPU time

Answer key 

34.0.101 NIELIT Scientific Assistant June 2025 | Question: 110



110. Which of the following is a characteristic of Boyce-Codd Normal Form (BCNF) ?

(A) Every determinant is a candidate key

(B) There are no transitive dependencies

(C) There are no partial dependencies

(D) All of the above

110. Which of the following is a characteristic of Boyce-Codd Normal Form (BCNF) ?

- A. Every determinant is a candidate key
- B. There are no transitive dependencies
- C. There are no partial dependencies
- D. All of the above

34.0.102 NIELIT Scientific Assistant June 2025 | Question: 109



109. The Power Spectral Density (PSD) of a random signal is modified by an LTI system in what way ?
 (A) It is completely filtered out
 (B) It is multiplied by the square of the system's frequency response
 (C) It is unaffected
 (D) It is multiplied by the impulse response

109. The Power Spectral Density (PSD) of a random signal is modified by an LTI system in what way ?

- A. It is completely filtered out
- B. It is multiplied by the square of the system's frequency response
- C. It is unaffected
- D. It is multiplied by the impulse response

nielit-sta-2025

34.0.103 NIELIT Scientific Assistant June 2025 | Question: 108



108. Which type of cryptography uses the same key for both encryption and decryption ?
 (A) Digital signatures
 (B) Asymmetric key cryptography
 (C) Hash functions
 (D) Symmetric key cryptography

108. Which type of cryptography uses the same key for both encryption and decryption ?

- (A) Digital signatures
- (B) Asymmetric key cryptography
- (C) Hash functions
- (D) Symmetric key cryptography

nielit-sta-2025

34.0.104 NIELIT Scientific Assistant June 2025 | Question: 107



107. Which of the following operations can be performed in $O(\log n)$ time in a binary heap ?
 (A) Deleting the root element
 (B) Inserting an element
 (C) Both (A) and (B)
 (D) Searching for an element

107. Which of the following operations can be performed in $O(\log n)$ time in a binary heap ?

- (A) Deleting the root element
- (B) Inserting an element
- (C) Both (A) and (B)
- (D) Searching for an element

nielit-sta-2025

34.0.105 NIELIT Scientific Assistant June 2025 | Question: 106



106. Which of the following statements is true for a relation in 3rd Normal Form (3NF) ?
 (A) It is in 2nd Normal Form (2NF)
 (B) It does not contain transitive dependencies
 (C) It does not contain partial dependencies
 (D) All of the above

106. Which of the following statements is true for a relation in 3rd Normal Form (3NF) ?

- A. It is in 2nd Normal Form (2NF)
- B. It does not contain transitive dependencies
- C. It does not contain partial dependencies
- D. All of the above

dependencies

nielit-sta-2025

Answer key 

34.0.106 NIELIT Scientific Assistant June 2025 | Question: 105



105. The number of bits used for quantization in PCM determines :
(A) The transmission speed
(B) The power efficiency
(C) The signal-to-noise ratio (SNR)
(D) The bandwidth of the system

105. The number of bits used for quantization in PCM determines:

- | | |
|------------------------------------|--------------------------------|
| A. The transmission speed | B. The power efficiency |
| C. The signal-to-noise ratio (SNR) | D. The bandwidth of the system |

nielit-sta-2025

34.0.107 NIELIT Scientific Assistant June 2025 | Question: 104



104. Which of the following operations does a floating-point representation support that a fixed-point representation cannot ?
(A) Arithmetic operations on integers
(B) Operations on very large or very small numbers
(C) Simple addition and subtraction
(D) Operations on binary fractions

104. Which of the following operations does a floating-point representation support that a fixed-point representation cannot?

- | | |
|--------------------------------------|---|
| A. Arithmetic operations on integers | B. Operations on very large or very small numbers |
| C. Simple addition and subtraction | D. Operations on binary fractions |

nielit-sta-2025

34.0.108 NIELIT Scientific Assistant June 2025 | Question: 103



103. Which of the following is a property of a regular language ?
(A) It cannot be represented by a finite automaton.
(B) It cannot be represented by a regular expression.
(C) It can only be represented by a context-free grammar.
(D) It is closed under union, concatenation, and Kleene star operations.

103. Which of the following is a property of a regular language ?

- A. It cannot be represented by a finite automaton.
B. It cannot be represented by a regular expression.
C. It can only be represented by a context-free grammar.
D. It is closed under union, concatenation, and Kleene star operations.

nielit-sta-2025

Answer key 

34.0.109 NIELIT Scientific Assistant June 2025 | Question: 102



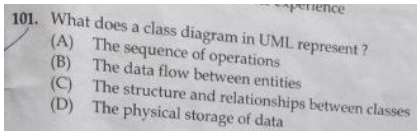
102. Which of the following is true about Turing Machines ?
(A) Turing machines are equivalent to finite automata
(B) Turing machines can simulate any computation that can be described algorithmically
(C) Turing machines are not capable of recognizing context-free languages
(D) Turing machines cannot halt

102. Which of the following is true about Turing Machines ?

- A. Turing machines are equivalent to finite automata
- B. Turing machines can simulate any computation that can be described algorithmically
- C. Turing machines are not capable of recognizing context-free languages
- D. Turing machines cannot halt

nielit-sta-2025

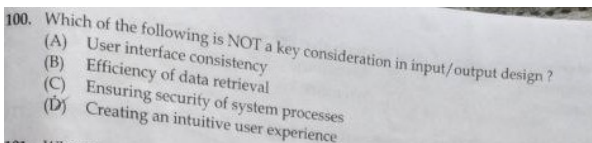
34.0.110 NIELIT Scientific Assistant June 2025 | Question: 101



101. What does a class diagram in UML represent ?
- (A) The sequence of operations
 - (B) The data flow between entities
 - (C) The structure and relationships between classes
 - (D) The physical storage of data

nielit-sta-2025

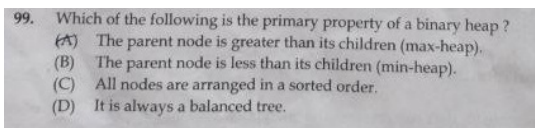
34.0.111 NIELIT Scientific Assistant June 2025 | Question: 100



100. Which of the following is NOT a key consideration in input/output design ?
- A. User interface consistency
 - B. Efficiency of data retrieval
 - C. Ensuring security of system processes
 - D. Creating an intuitive user experience

nielit-sta-2025

34.0.112 NIELIT Scientific Assistant June 2025 | Question: 99



99. Which of the following is the primary property of a binary heap ?
- A. The parent node is greater than its children (max-heap).
 - B. The parent node is less than its children (min-heap).
 - C. All nodes are arranged in a sorted order.
 - D. It is always a balanced tree.

nielit-sta-2025

34.0.113 NIELIT Scientific Assistant June 2025 | Question: 98



98. Which of the following is an example of a combinational circuit ?
(A) Counter (B) Register (C) Multiplexer (D) Shift Register

98. Which of the following is an example of a combinational circuit ?

- A. Counter B. Register C. Multiplexer D. Shift Register

nielit-sta-2025

Answer key

34.0.114 NIELIT Scientific Assistant June 2025 | Question: 97



97. Which of the following is the main objective of memory management in an operating system ?
(A) Efficient allocation of memory (B) Protection of memory
(C) Virtual memory management (D) All of the options

97. Which of the following is the main objective of memory management in an operating system ?

- A. Efficient allocation of memory B. Protection of memory
C. Virtual memory management D. All of the options

nielit-sta-2025

Answer key

34.0.115 NIELIT Scientific Assistant June 2025 | Question: 96



96. What is the primary use of a Karnaugh Map (K-map) in Boolean algebra ?
(A) To simplify logic circuits
(B) To find the truth table
(C) To convert a Boolean expression into its canonical form
(D) To convert between logic gates

96. What is the primary use of a Karnaugh Map (K-map) in Boolean algebra ?

- A. To simplify logic circuits B. To find the truth table
C. To convert a Boolean expression into its canonical form D. To convert between logic gates

nielit-sta-2025

Answer key

34.0.116 NIELIT Scientific Assistant June 2025 | Question: 95



95. Which of the following is NOT a component of syntax analysis in a compiler ?
(A) Syntax tree (B) Parse tree (C) Token (D) Grammar rules

95. Which of the following is NOT a component of syntax analysis in a compiler ?

- A. Syntax tree B. Parse tree C. Token D. Grammar rules

nielit-sta-2025

34.0.117 NIELIT Scientific Assistant June 2025 | Question: 94



94. What is the main advantage of a Link-State Routing algorithm over a Distance-Vector Routing algorithm ?
- (A) It is more efficient in terms of bandwidth usage
 - (B) It has a faster convergence time
 - (C) It requires less memory
 - (D) It is simpler to implement

94. What is the main advantage of a Link-State Routing algorithm over a Distance-Vector Routing algorithm?

- A. It is more efficient in terms of bandwidth usage
- B. It has a faster convergence time
- C. It requires less memory
- D. It is simpler to implement

nielit-sta-2025

Answer key

34.0.118 NIELIT Scientific Assistant June 2025 | Question: 93



93. What is the main purpose of a file system ?
- (A) To protect the system from malware
 - (B) To manage the data stored on a storage device
 - (C) To provide networking services
 - (D) To execute user programs

93. What is the main purpose of a file system ?

- A. To protect the system from malware
- B. To manage the data stored on a storage device
- C. To provide networking services
- D. To execute user programs

nielit-sta-2025

Answer key

34.0.119 NIELIT Scientific Assistant June 2025 | Question: 92



92. What happens in TCP when congestion is detected in the network ?
- (A) The sender reduces its congestion window
 - (B) The sender increases its congestion window
 - (C) The receiver sends an acknowledgment for all unacknowledged packets
 - (D) The sender stops sending data

92. What happens in TCP when congestion is detected in the network ?

- A. The sender reduces its congestion window
- B. The sender increases its congestion window
- C. The receiver sends an acknowledgment for all unacknowledged packets
- D. The sender stops sending data

nielit-sta-2025

34.0.120 NIELIT Scientific Assistant June 2025 | Question: 91



91. Which of the following is an intermediate representation used by compilers ?
- (A) Abstract syntax tree
 - (B) Parse tree
 - (C) Machine code
 - (D) Source code

91. Which of the following is an intermediate representation used by compilers ?

- A. Abstract syntax tree
- B. Parse tree
- C. Machine code
- D. Source code

nielit-sta-2025

34.1

General Awareness (4)

34.1.1 General Awareness: NIELIT 2022 April Scientist B | Section A | Question: 23

Assertion (A): Increase in carbon dioxide would melt polar ice.

Reason (R): Global temperature would rise.



- A. Both (A) and (R) are true but (R) is not the correct explanation of (A)
- B. Both (A) and (R) are true and (R) is the correct explanation of (A)
- C. (A) is true but (R) is false.
- D. (A) is false but (R) is true.

nielit2022apr-scientistb general-awareness logical-reasoning

34.1.2 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 135

The Ohio State Leadership Studies revealed _____ and initiating structure as two major dimensions of leadership behaviour:



- | | |
|------------------|------------------|
| A. Control | B. Communication |
| C. Collaboration | D. Consideration |

nielit2023feb-scientistc general-awareness

34.1.3 General Awareness: NIELIT 2023 Feb Scientist C | Section E | Question: 144

Which style of leadership focuses on goals, standards, and organization?



- | | |
|------------------------|--------------------------------|
| A. Task leadership | B. Social leadership |
| C. Semantic leadership | D. Transformational leadership |

nielit2023feb-scientistc general-awareness

34.1.4 General Awareness: NIELIT 2023 Feb Scientist D | Section A | Question: 24

Directions: Answer questions (24-25) on the basis of following instructions :

Instructions: For the Assertions (A) and Reason (R) below, choose the correct alternative from the following :



- (I) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (II) Both (A) and (R) are true and (R) is not the correct explanation of (A).
- (III) (A) is true but (R) is false.
- (IV) (A) is false but (R) is true.

Assertion (A): The filament in an electric bulb does not burn up although its temperature

is about 2700°C when it glows.

Reason (R): Bulb filament is made of tungsten.

- A. (I) B. (II) C. (III) D. (IV)

nielit2023feb-scientistd logical-reasoning general-awareness analytical-aptitude

34.2 Is&software Engineering (14)

34.2.1 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 132



You are asked by your instructor to discuss about Contingency theory. What should you be included in your discussion?

- A. It is a theory in leadership that views the pattern of leader behaviour as dependent on the interaction of the personality and the needs of the situation.
- B. It emphasizes that both leaders and followers should act on one another to raise their motivation.
- C. It states that leadership qualities inspire followers to be motivated by what they do.
- D. None of the options

nielit2023feb-scientistc is&software-engineering logical-reasoning

34.2.2 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 137



A Charismatic leader's _____ is the key to follower acceptance.

- A. Energy B. History with the organization
- C. Credentials D. Vision

nielit2023feb-scientistc is&software-engineering

34.2.3 Is&software Engineering: NIELIT 2023 Feb Scientist C | Section E | Question: 140



Participative leadership has which of the following characteristics?

- A. Believe success arises from leaders and staff working together
- B. Employs a clear chain of command
- C. Takes the view that rewards and punishment motivate staff
- D. Seeks to involve staff in the decision making process

nielit2023feb-scientistc is&software-engineering logical-reasoning

34.2.4 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 123



This style is a one side approach where the leader-manager pays much attention to the security and comfort of the employees but he has low concern for production.

- A. Impoverished
- B. Country club
- C. Dictatorial or Perish
- D. The Team or Sound

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.5 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 124



_____ are the theories which gives an idea about what employees wants or needs and what are the key factors the managers can utilize to motivate the employees.

- A. Maslow theory
- B. Herzberg Theory
- C. Process Theory
- D. Content Theory

nielit2023feb-scientistd is&software-engineering

Answer key 

34.2.6 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 125



In addition to basic managerial functions of planning, organizing, staffing, directing, and controlling, leaders are ascribed :

- A. Procedural and external roles.
- B. Procedural and internal roles.
- C. Strategic and internal roles.
- D. Strategic and external roles.

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.7 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 128



_____ commitment refers to an employee's obligation to remain with an organization for moral or ethical reasons.

- A. Affective
- B. Continuance
- C. Theoretical
- D. Normative

nielit2023feb-scientistd is&software-engineering

34.2.8 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question:



The process by which an older and more experienced person helps to socialize and encourage younger organizational colleagues is called _____.

- A. Evaluating B. Consulting C. Mentoring D. Networking

nielit2023feb-scientistd is&software-engineering

34.2.9 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 135



In managerial grid, the style of management depicts the style of a leader who is neither concerned about the people nor does he care about the tasks to be performed is _____.

- A. task management B. impoverished style
C. country club D. team management style

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.10 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 136



A range of management behaviors based on the various ways that task-oriented and, employee-oriented styles (each expressed as a continuum on a scale of 1 to 9) is known as :

- A. Bi-dimensional managerial grid B. Rectangular managerial grid
C. Diagonal managerial grid D. BCG Matrix

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.11 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 138



The trait approach to leadership suggests that:

- A. Leaders have special innate qualities. B. Leadership traits are clearly visible.
C. Traits are based on social class. D. Traits cannot be measured.

nielit2023feb-scientistd is&software-engineering

34.2.12 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 143



All decision-making power is centralized in the leader is under _____.

- A. autocratic style B. liberal leader

C. democratic leader

D. institutional leader

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.13 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 147



Which path-goal leadership style leads to greater satisfaction when tasks are ambiguous or stressful?

A. Directive

B. Supportive

C. Participative

D. Mixed

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.2.14 Is&software Engineering: NIELIT 2023 Feb Scientist D | Section E | Question: 148



Which role focuses on bringing about order and consistency by drawing up formal plans?

A. Leadership

B. Management

C. Task structure

D. Initiating structure

nielit2023feb-scientistd is&software-engineering logical-reasoning

34.3 Logical Reasoning (1)

34.3.1 Logical Reasoning: NIELIT 2023 Feb Scientist D | Section E | Question: 144



Groups created by managerial decision in order to accomplish stated goals of the organization are called _____.

A. Formal groups

B. Informal groups

C. Task groups

D. Interest groups

nielit2023feb-scientistd logical-reasoning

Answer Keys

34.0.1	D	34.0.2	C	34.0.3	A	34.0.4	B	34.0.5	B
34.0.6	B	34.0.7	C	34.0.8	A	34.0.9	B	34.0.10	B
34.0.11	D	34.0.12	C	34.0.13	B	34.0.14	A	34.0.15	B
34.0.16	D	34.0.17	D	34.0.18	B	34.0.19	B	34.0.20	C
34.0.21	C	34.0.22	C	34.0.23	A	34.0.24	C	34.0.25	A

34.0.26	D	34.0.27	A	34.0.28	B	34.0.29	D	34.0.30	C
34.0.31	B	34.0.32	D	34.0.33	D	34.0.34	C	34.0.35	A
34.0.36	D	34.0.37	D	34.0.38	C	34.0.39	B	34.0.40	C
34.0.41	C	34.0.42	B	34.0.43	D	34.0.44	B	34.0.45	A
34.0.46	D	34.0.47	B	34.0.48	A	34.0.49	D	34.0.50	D
34.0.51	C	34.0.52	C	34.0.53	A	34.0.54	A	34.0.55	B
34.0.56	D	34.0.57	C	34.0.58	A	34.0.59	A	34.0.60	D
34.0.61	C	34.0.62	B	34.0.63	C	34.0.64	A	34.0.65	A
34.0.66	C	34.0.67	A	34.0.68	D	34.0.69	D	34.0.70	B
34.0.71	C	34.0.72	B	34.0.73	D	34.0.74	A	34.0.75	A
34.0.76	A	34.0.77	C	34.0.78	C	34.0.79	B	34.0.80	A
34.0.81	D	34.0.82	C	34.0.83	D	34.0.84	B	34.0.85	A
34.0.86	B	34.0.87	A	34.0.88	B	34.0.89	A	34.0.90	D
34.0.91	B	34.0.92	D	34.0.93	B	34.0.94	C	34.0.95	C
34.0.96	B	34.0.97	D	34.0.98	C	34.0.99	A	34.0.100	D
34.0.101	A	34.0.102	B	34.0.103	D	34.0.104	C	34.0.105	D
34.0.106	C	34.0.107	B	34.0.108	D	34.0.109	B	34.0.110	C
34.0.111	C	34.0.112	D	34.0.113	C	34.0.114	D	34.0.115	A
34.0.116	C	34.0.117	B	34.0.118	B	34.0.119	A	34.0.120	A
34.1.1	A	34.1.2	D	34.1.3	A	34.1.4	A	34.2.1	A
34.2.2	D	34.2.3	D	34.2.4	B	34.2.5	D	34.2.6	D
34.2.7	D	34.2.8	C	34.2.9	B	34.2.10	A	34.2.11	A
34.2.12	A	34.2.13	B	34.2.14	B	34.3.1	A		